`frob_roots`: -14.6 % (m = 12) and -18.5 % (m = 14) counted S in the char-2 D4 index calculus

Protocol: PROTOCOL.md, committed (d2eec8df3) before any session.

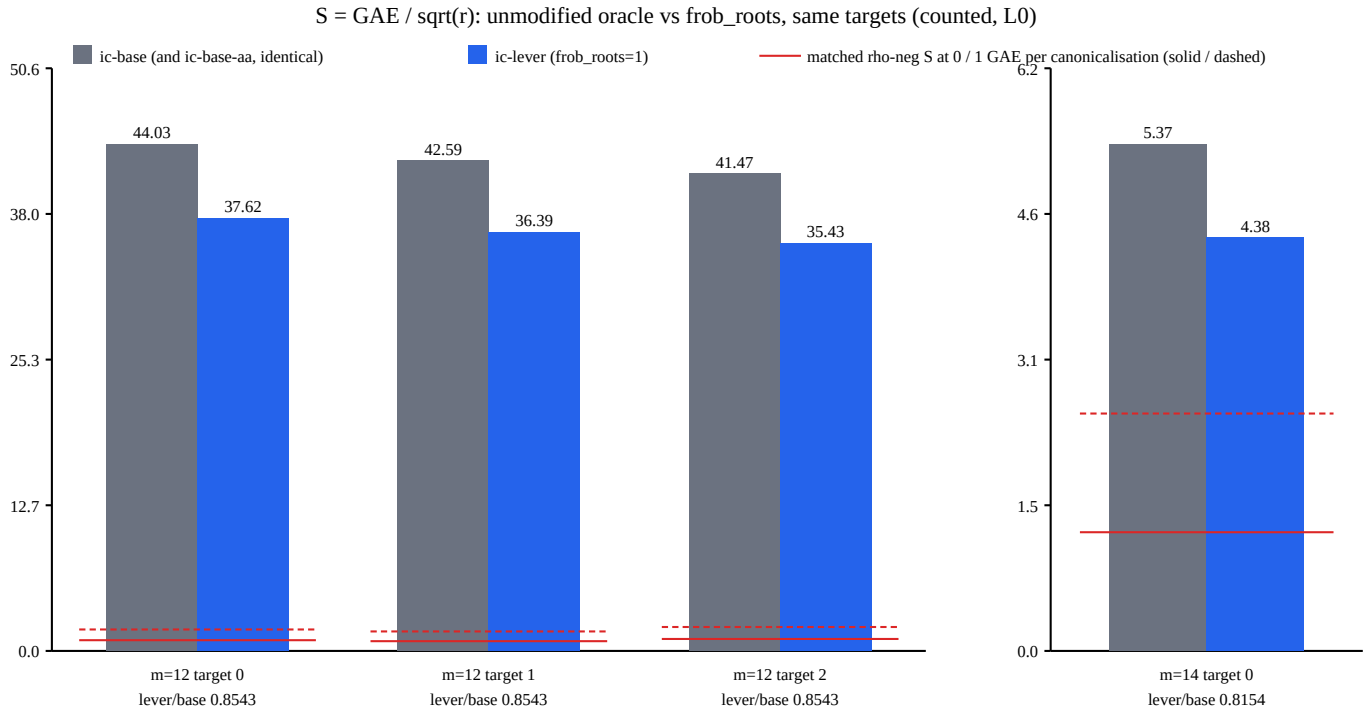
Sessions: `results/lever-m12` and `results/lever-m14` (L0, `--cpus none --allow-busy`, `RAYON_NUM_THREADS=1`). **Class: engineering.** Counted-unit result only; no wall-clock claim.

What changed

`src/cryptanalysis/gaudry_quartic_bin.rs`: `SqMod` and `roots_with(.., frob)`. The `X^q - X mod p` and Cantor-Zassenhaus squarings use `(sum a_i X^i)^2 = sum a_i^2 X^(2i)` with a per-modulus table of `X^(d+j) mod p`, instead of a generic product plus reduction. Oracle parameter `frob_roots=1` (`BinOptions.frob_roots`, default off). Identical polynomial, roots, rational solutions, decompositions, witnesses and trial sequence.

The other two candidates were not available (PROTOCOL.md): the first-dependency stop is already what `incremental-gauss` does here (linear algebra is 0.1 % of S); no exact cheap rejection test exists because the solve cost does not depend on the outcome.

Result (paired, same instances, targets and seeds; every run `verified`)



curve	arm	S per target	trials (identical across arms)
m = 12	ic-base	44.03, 42.59, 41.47	186,121; 180,033; 175,246
m = 12	ic-lever	37.62, 36.39, 35.43	same
m = 12	ic-base-aa	44.03, 42.59, 41.47	same
m = 14	ic-base	5.3701	749,350
m = 14	ic-lever	4.3786	749,350
m = 14	ic-base-aa	5.3701	749,350

`ic-base` reproduces the 2026-10-15 sessions exactly (44.0338 at m = 12 target 0, 5.3701 at m = 14).

paired S ratio lever / base	value	95 % interval (`ecbench compare`)	A/A (base-aa / base)
m = 12, 3 targets	0.8543	[0.8543, 0.8543] (cluster)	1.0000 [1.000, 1.000]
m = 14, 1 target	0.8154	[0.815, 0.815] (within; one target)	1.0000

IC arms emit no `*_uncharged` counter (`unpriced: []`, `lower_bound: false`), so the lever/base ratio is the same at w = 0, 0.1 and 1 GAE per unpriced native operation: 0.8543 at all three weights at

m = 12 and 0.8154 at all three at m = 14. Intervals are degenerate because counts are deterministic; per-target ratios at m = 12 are 0.85430, 0.85430, 0.85432. m = 14 is one target: `within` says nothing about other targets. Caveat: non-multiplication native work in the eigen step (XOR accumulation, table lookups) is not counted for either IC arm; the lever removes it roughly in proportion, but that is not measured.

Per-trial components at m = 12 (300 random targets, `examples/d4_char2_profile`, F_q mults): root finding 55,505 -> 22,977; total 224,196 -> 191,435; charpoly 87,728, echelon 54,809, normal forms 21,913 unchanged (asserted equal in the unit test). In-process wall per trial 4.595 ms -> 4.439 ms (ratio 0.966; shared VM, one run, descriptive only).

Is this relabelling or accounting?

No. Trials, relations, rank and every other component are identical; no work moved to another phase. The removed multiplications are executed table multiplications of a generic product (about 64^2 per squaring) replaced by 64 squarings, and the counter is unchanged. Limit: counted S falls 14.6 % at m = 12 but oracle wall time only about 3 % in the one in-process comparison, because the removed multiplications are cheap lookups next to the dense linear-algebra loops. This is a counted-unit improvement; a wall-clock improvement is not established (needs L2).

Distance to the matched rho (`rho-neg`, same targets, same session)

Rho carries `canonicalisations_uncharged` (about one per step, 4,945,607 at m = 12 target 0), priced at w GAE each on rho. IC / rho ratios, base -> lever:

curve	w = 0	w = 0.1	w = 1
m = 12 t0	47.2 -> 40.4	42.9 -> 36.7	23.6 -> 20.2
m = 12 t1	50.3 -> 43.0	45.7 -> 39.1	25.1 -> 21.5
m = 12 t2	39.9 -> 34.1	36.3 -> 31.0	19.9 -> 17.0
m = 14 t0	4.27 -> 3.48	3.88 -> 3.16	2.13 -> 1.74

Rho is ahead at every weight (order holds across the band). Against rho's theoretical cost the m = 14 ratio is 4.94, down from 6.06. Distance remaining, not a beat of rho.

Controls

- A/A: `ic-base-aa` equals `ic-base` to the last digit on every target.

- Positive control: the profile's independent `roots` component falls 59 % (55,505 -> 22,977).

- Tests (`cargo test --release --lib gaudry_quartic_bin`, 7 pass): roots equal the generic path for degrees 2-64, m = 3-12; solutions and decompositions equal over 12 solves.

Replay audit

`ecbench verify --replay-all` (binary sha256 13818f3e..., the binary that measured both sessions; auditor on the same host class, so this is a same-host replay certificate, not an independent-host one):

session	records	verified	replays reproduced	problems	audit receipt SHA-256
`results/lever-m12`	12	12	12	0	781a3d8500e6093982f7a2a9da5057d255402ea96a4972f33186f5151967034b
`results/lever-m14`	4	4	4	0	3d63173bdd224b9c85e8e02bce8d98fe00625b17f6896ebea151a62b5346689d

Receipts: `results/lever-m12.audit.json`, `results/lever-m14.audit.json`.

Next

1. Charpoly (42 %): 65 dense determinants per trial; structure-exploiting determinant or Krylov/Berlekamp-Massey with an exact fallback.

2. Echelon + normal forms (37 %): share elimination across trials through the t_R dependence.

3. Raise the 0.5 % per-trial success (second coset, larger Artin-Schreier class).

4. An L2 wall session to see whether the counted gain survives.

Not touched: leaderboard, `benchmarks/README.md` (no VALIDATION.json), no PR. Graph:

figures/s_lever.svg (data results/table-m*.tsv, layout figures/plot.py); PDF: REPORT.pdf.